Uncertainty Propagation for a Ratio of GP Derivatives: From Posterior Covariance to Parameter Confidence Intervals

Abstract

We work through a concrete example of how Gaussian-process (GP) posterior uncertainty propagates to an ODE parameter that is determined by a ratio of derivatives. Suppose an observable $g(t)$ is fitted by a GP, and a parameter satisfies the algebraic relation

$$a = \frac{g''(t_*)}{g'''(t_*)}.$$

We show how to compute $\mathrm{Var}(a)$ from the GP posterior, starting from the kernel and ending at a closed-form expression. The calculation illustrates the general IFT/GLS pipeline of the `ODEParameterEstimation.jl` package in a setting simple enough to follow every step.

Contents

1 Setup: One Observable, One Parameter **1**

2 Step 1: GP Posterior for Function Values **1**
2.1 Standard GP conditioning . . . 1
2.2 Key observation . . . 2

3 Step 2: GP Posterior for Derivatives **2**
3.1 Derivatives of a GP are a GP . . . 2
3.2 SE kernel derivatives via Hermite polynomials . . . 2
3.3 Prior covariance at the same point ($\Delta t = 0$, i.e. $u = 0$) . . . 3
3.4 Posterior covariance of the derivative vector . . . 3
3.5 Why the posterior cross-covariance $\mathrm{Cov}(g'', g''')$ is nonzero . . . 4

4 Step 3: Extracting the Relevant 2×2 Block **4**

5 Step 4: Delta Method for the Ratio **5**
5.1 The general delta method . . . 5
5.2 Application to $h(u, v) = u/v$ . . . 5
5.3 Expanding the variance . . . 5
5.4 Interpreting the three terms . . . 5

6 Step 5: Connection to the IFT/GLS Framework **5**
6.1 Rewriting the constraint . . . 6
6.2 Jacobians . . . 6
6.3 Sensitivity matrix . . . 6
6.4 Parameter covariance . . . 6
6.5 The GLS formula for overdetermined systems . . . 6

7 Step 6: Numerical Scaling — Why the Denominator Hurts **6**
7.1 Prior variance growth with derivative order . . . 6
7.2 Posterior variance is smaller, but follows the same trend . . . 7
7.3 Consequence for the ratio . . . 7

8 Step 7: A Complete Numerical Example **7**
8.1 Setting . . . 7
8.2 Computing Var(a) . . . 8
8.3 Breakdown by term . . . 8

9 Summary **8**

1 Setup: One Observable, One Parameter

We observe noisy samples of a smooth function g:

$$y_i = g(t_i) + \varepsilon_i, \qquad \varepsilon_i \sim \mathcal{N}(0, \sigma_n^2), \qquad i = 1, \ldots, N.$$

We fit a GP with squared-exponential (SE) kernel

$$k(\Delta t) = \sigma^2 \exp\left(-\frac{\Delta t^2}{2\ell^2}\right), \qquad \Delta t = t - t'.$$

After conditioning on the data, the GP posterior gives us not just $g(t_*)$ but also all derivatives $g'(t_*), g''(t_*), g'''(t_*), \ldots$ as jointly Gaussian random variables.

The ODE model tells us that a parameter a satisfies

$$a = \frac{g''(t_*)}{g'''(t_*)}.$$

Our goal: compute Var(a) as a function of the GP posterior.

2 Step 1: GP Posterior for Function Values

2.1 Standard GP conditioning

Define the training kernel matrix

$$K \in \mathbb{R}^{N \times N}, \qquad K_{ij} = k(t_i - t_j),$$

and the noisy kernel matrix

$$\tilde{K} = K + \sigma_n^2 I.$$

Let $\mathbf{y} = (y_1, \ldots, y_N)^\top$ be the observed data vector.

At any test point t_*, the posterior of the latent function value $g(t_*)$ is Gaussian:

$$\mu(t_*) = \mathbf{k}_*^\top \tilde{K}^{-1} \mathbf{y}, \tag{1}$$

$$\sigma^2_{\text{post}}(t_*) = k(0) - \mathbf{k}_*^\top \tilde{K}^{-1} \mathbf{k}_*, \tag{2}$$

where

$$\mathbf{k}_* = \begin{bmatrix} k(t_* - t_1) \\ \vdots \\ k(t_* - t_N) \end{bmatrix} \in \mathbb{R}^N.$$

2.2 Key observation

The posterior *mean* (1) depends on the data values $\mathbf{y}$.

The posterior *variance* (2) depends only on the kernel hyperparameters $(\sigma^2, \ell, \sigma_n^2)$ and the training point locations $\{t_i\}$—not on $\mathbf{y}$ itself.

However, in practice we estimate the hyperparameters from the data by maximizing the marginal likelihood, so the variance indirectly depends on $\mathbf{y}$ through $(\hat{\sigma}^2, \hat{\ell}, \hat{\sigma}_n^2)$.

3 Step 2: GP Posterior for Derivatives

3.1 Derivatives of a GP are a GP

Since the SE kernel is infinitely differentiable, the GP posterior for any derivative $g^{(p)}(t_*)$ is also Gaussian. The key identity is:

$$\mathrm{Cov}\big(g^{(p)}(t),\ g^{(q)}(t')\big) = \frac{\partial^{p+q}}{\partial t^p\, \partial t'^{\,q}}\, k(t, t').$$

We denote this derivative kernel as

$$k^{(p,q)}(t, t') = \frac{\partial^{p+q}}{\partial t^p\, \partial t'^{\,q}}\, k(t, t').$$

3.2 SE kernel derivatives via Hermite polynomials

For the SE kernel with normalized distance $u = \Delta t/\ell$, we have

$$k(\Delta t) = \sigma^2 e^{-u^2/2}.$$

The n-th derivative of $e^{-u^2/2}$ with respect to u introduces the probabilist's Hermite polynomial:

$$\frac{d^n}{du^n} e^{-u^2/2} = (-1)^n\, \mathrm{He}_n(u)\, e^{-u^2/2},$$

where the Hermite polynomials satisfy the recurrence

$$\mathrm{He}_0(u) = 1, \qquad \mathrm{He}_1(u) = u, \qquad \mathrm{He}_{n+1}(u) = u\,\mathrm{He}_n(u) - n\,\mathrm{He}_{n-1}(u).$$

Accounting for the chain rule ($\partial/\partial t = (1/\ell)\,\partial/\partial u$ and $\partial/\partial t' = -(1/\ell)\,\partial/\partial u$ since $u = (t - t')/\ell$), the general formula is

$$k^{(p,q)}(\Delta t) = (-1)^p \cdot \frac{\sigma^2}{\ell^{p+q}} \cdot \mathrm{He}_{p+q}(u) \cdot e^{-u^2/2}. \tag{3}$$

For reference, the first several Hermite polynomials are:

n	$\mathrm{He}_n(u)$
0	1
1	u
2	$u^2 - 1$
3	$u^3 - 3u$
4	$u^4 - 6u^2 + 3$
5	$u^5 - 10u^3 + 15u$
6	$u^6 - 15u^4 + 45u^2 - 15$

3.3 Prior covariance at the same point ($\Delta t = 0$, i.e. $u = 0$)

At $\Delta t = 0$:

$$k^{(p,q)}(0) = (-1)^p \cdot \frac{\sigma^2}{\ell^{p+q}} \cdot \mathrm{He}_{p+q}(0).$$

Since $\mathrm{He}_n(0) = 0$ for odd n and $\mathrm{He}_n(0) = (-1)^{n/2}(n-1)!!$ for even n, we get:

- If $p+q$ is odd: $k^{(p,q)}(0) = 0$. (Prior covariance vanishes by symmetry.)
- If $p+q$ is even: $k^{(p,q)}(0) = (-1)^p \cdot \dfrac{\sigma^2}{\ell^{p+q}} \cdot (-1)^{(p+q)/2} \cdot (p+q-1)!!$

Concretely, the prior covariance matrix for $[g, g', g'', g''']$ at a single point is:

$$\Sigma_{\text{prior}} = \sigma^2 \begin{bmatrix} 1 & 0 & -\ell^{-2} & 0 \\ 0 & \ell^{-2} & 0 & -3\ell^{-4} \\ -\ell^{-2} & 0 & 3\ell^{-4} & 0 \\ 0 & -3\ell^{-4} & 0 & 15\ell^{-6} \end{bmatrix}.$$

Important observation: Under the prior, $\mathrm{Cov}(g'', g''') = 0$ because $2 + 3 = 5$ is odd. This will *not* remain zero after conditioning on data.

3.4 Posterior covariance of the derivative vector

Define the derivative vector at test point t_*:

$$\mathbf{f}_* = \begin{bmatrix} g(t_*) \\ g'(t_*) \\ g''(t_*) \\ g'''(t_*) \end{bmatrix}.$$

The cross-covariance between $\mathbf{f}_*$ and the training function values $\mathbf{g} = (g(t_1), \ldots, g(t_N))^\top$ is:

$$K_{*n} \in \mathbb{R}^{4\times N}, \qquad (K_{*n})_{p+1,\,j} = k^{(p,0)}(t_* - t_j) = \frac{\partial^p}{\partial t^p} k(t_* - t_j),$$

for $p = 0, 1, 2, 3$ and $j = 1, \ldots, N$.

The prior covariance of $\mathbf{f}_*$ is $\Sigma_{\text{prior}} \in \mathbb{R}^{4\times 4}$ as above.

The GP posterior of $\mathbf{f}_*$ given data $\mathbf{y}$ is:

$$\boldsymbol{\mu}_* = K_{*n}\, \tilde{K}^{-1}\, \mathbf{y}, \tag{4}$$

$$\Sigma_{\text{post}} = \Sigma_{\text{prior}} - K_{*n}\, \tilde{K}^{-1}\, K_{*n}^\top. \tag{5}$$

Equation (4) says: *the posterior mean of each derivative is a weighted combination of the observed data*, with weights derived from the kernel derivatives.

Equation (5) says: *conditioning on data reduces uncertainty from the prior.* The reduction is encoded in the $K_{*n}\, \tilde{K}^{-1}\, K_{*n}^\top$ term. This term depends on how informative the training points are about the derivatives at t_*.

3.5 Why the posterior cross-covariance $\mathrm{Cov}(g'', g''')$ is nonzero

Even though the prior gives $\mathrm{Cov}_{\text{prior}}(g'', g''') = 0$, the data-reduction term

$$\big(K_{*n}\,\tilde{K}^{-1}\,K_{*n}^{\top}\big)_{3,4}$$

is generically nonzero. Intuitively, the training data near t_* provide information that correlates even- and odd-order derivatives: observing g at asymmetrically placed points around t_* breaks the symmetry that made the prior cross-covariance vanish.

4 Step 3: Extracting the Relevant 2×2 Block

For the ratio $a = g''(t_*)/g'''(t_*)$, we only need the subvector

$$\mathbf{z} = \begin{bmatrix} g''(t_*) \\ g'''(t_*) \end{bmatrix} = \begin{bmatrix} u \\ v \end{bmatrix},$$

which has posterior distribution

$$\mathbf{z} \sim \mathcal{N}\left(\begin{bmatrix} \mu_u \\ \mu_v \end{bmatrix},\ \Sigma_z = \begin{bmatrix} \sigma_{uu} & \sigma_{uv} \\ \sigma_{uv} & \sigma_{vv} \end{bmatrix} \right),$$

where Σ_z is the (3:4, 3:4) subblock of Σ_{post} from (5).

Explicitly:

$$\sigma_{uu} = \underbrace{k^{(2,2)}(0)}_{\text{prior}} - \sum_{i,j} k^{(2,0)}(t_* - t_i)\,[\tilde{K}^{-1}]_{ij}\,k^{(2,0)}(t_* - t_j), \tag{6}$$

$$\sigma_{vv} = k^{(3,3)}(0) - \sum_{i,j} k^{(3,0)}(t_* - t_i)\,[\tilde{K}^{-1}]_{ij}\,k^{(3,0)}(t_* - t_j), \tag{7}$$

$$\sigma_{uv} = \underbrace{k^{(2,3)}(0)}_{=0\ (\text{prior})} - \sum_{i,j} k^{(2,0)}(t_* - t_i)\,[\tilde{K}^{-1}]_{ij}\,k^{(3,0)}(t_* - t_j). \tag{8}$$

The prior term in (8) vanishes (odd total order), but the data-reduction sum does not.

5 Step 4: Delta Method for the Ratio

5.1 The general delta method

If $\mathbf{z} \sim \mathcal{N}(\boldsymbol{\mu}, \Sigma_z)$ and $h(\mathbf{z})$ is a smooth scalar function, then to first order:

$$h(\mathbf{z}) \approx h(\boldsymbol{\mu}) + \nabla h(\boldsymbol{\mu})^{\top}(\mathbf{z} - \boldsymbol{\mu}),$$

and therefore

$$\mathrm{Var}\big(h(\mathbf{z})\big) \approx \nabla h(\boldsymbol{\mu})^{\top}\,\Sigma_z\,\nabla h(\boldsymbol{\mu}).$$

5.2 Application to $h(u, v) = u/v$

The partial derivatives are:

$$\frac{\partial h}{\partial u} = \frac{1}{v}, \qquad \frac{\partial h}{\partial v} = -\frac{u}{v^2}.$$

Evaluated at the posterior means:

$$\nabla h\big|_{\boldsymbol{\mu}} = \begin{bmatrix} 1/\mu_v \\ -\mu_u/\mu_v^2 \end{bmatrix}.$$

5.3 Expanding the variance

$$\begin{aligned}\mathrm{Var}(a) &\approx \nabla h^\top \Sigma_z \, \nabla h \\ &= \frac{1}{\mu_v^2}\,\sigma_{uu} \;-\; \frac{2\mu_u}{\mu_v^3}\,\sigma_{uv} \;+\; \frac{\mu_u^2}{\mu_v^4}\,\sigma_{vv}. \end{aligned} \tag{9}$$

Equivalently, factoring out $\hat{a}^2 = \mu_u^2/\mu_v^2$:

$$\boxed{\mathrm{Var}(a) \approx \hat{a}^2 \left[\frac{\sigma_{uu}}{\mu_u^2} \;-\; 2\rho\,\frac{\sqrt{\sigma_{uu}}}{\mu_u}\,\frac{\sqrt{\sigma_{vv}}}{\mu_v} \;+\; \frac{\sigma_{vv}}{\mu_v^2}\right] = \hat{a}^2\left[\mathrm{CV}_u^2 - 2\rho\,\mathrm{CV}_u\,\mathrm{CV}_v + \mathrm{CV}_v^2\right]} \tag{10}$$

where $\mathrm{CV}_u = \sqrt{\sigma_{uu}}/|\mu_u|$, $\mathrm{CV}_v = \sqrt{\sigma_{vv}}/|\mu_v|$, and $\rho = \sigma_{uv}/\sqrt{\sigma_{uu}\sigma_{vv}}$.

5.4 Interpreting the three terms

1. σ_{uu}/μ_v^2: uncertainty in the **numerator** g'', scaled by the denominator.
2. $\mu_u^2\sigma_{vv}/\mu_v^4$: uncertainty in the **denominator** g''', amplified by $1/\mu_v^2$. This is typically the dominant term because GP derivative variance grows rapidly with order.
3. $-2\mu_u\sigma_{uv}/\mu_v^3$: the **correlation correction**. If g'' and g''' are positively correlated (and $\hat{a} > 0$), this term is negative and *reduces* the variance. Intuitively, when numerator and denominator fluctuate together, the ratio is more stable.

6 Step 5: Connection to the IFT/GLS Framework

The `ODEParameterEstimation.jl` package does not compute $\mathrm{Var}(a)$ using the ratio formula directly. Instead, it uses the implicit function theorem (IFT), which gives the same answer through a different route. We show the equivalence.

6.1 Rewriting the constraint

Instead of $a = u/v$, write the constraint as

$$F(a, u, v) = a \cdot v - u = 0.$$

6.2 Jacobians

The constraint Jacobians are:

$$J_\theta = \frac{\partial F}{\partial a} = v = \mu_v \quad \text{(scalar, since one parameter)}, \tag{11}$$

$$J_z = \left[\frac{\partial F}{\partial u},\ \frac{\partial F}{\partial v}\right] = [-1,\ a] = \left[-1,\ \frac{\mu_u}{\mu_v}\right]. \tag{12}$$

6.3 Sensitivity matrix

The sensitivity of a to perturbations in the data is

$$S = -J_\theta^{-1}\,J_z = -\frac{1}{\mu_v}\left[-1,\ \mu_u/\mu_v\right] = \left[\frac{1}{\mu_v},\ -\frac{\mu_u}{\mu_v^2}\right].$$

This is exactly $\nabla h^\top$ from the delta method.

6.4 Parameter covariance

$$\Sigma_a = S\,\Sigma_z\,S^\top = \frac{\sigma_{uu}}{\mu_v^2} - \frac{2\mu_u\,\sigma_{uv}}{\mu_v^3} + \frac{\mu_u^2\,\sigma_{vv}}{\mu_v^4}.$$

This is identical to (9). The IFT and delta method are two views of the same linearization.

6.5 The GLS formula for overdetermined systems

In this example, we had exactly one constraint for one parameter (a square system), so J_θ^{-1} exists and the IFT formula applies directly.

In general, SIAN produces more equations than unknowns ($m > p$). The correct generalization is the GLS (generalized least squares) formula:

$$\Sigma_\theta = \left(J_\theta^\top\,\Sigma_z^{-1}\,J_\theta\right)^{-1}.$$

For the square case, this reduces to $J_\theta^{-1}\,\Sigma_z\,J_\theta^{-\top}$, recovering the IFT result.

7 Step 6: Numerical Scaling — Why the Denominator Hurts

7.1 Prior variance growth with derivative order

From the prior covariance matrix, the marginal prior variances are:

Quantity	Prior variance	Scales as
$g(t)$	σ^2	σ^2
$g'(t)$	σ^2/ℓ^2	$\sigma^2\ell^{-2}$
$g''(t)$	$3\sigma^2/\ell^4$	$\sigma^2\ell^{-4}$
$g'''(t)$	$15\sigma^2/\ell^6$	$\sigma^2\ell^{-6}$
$g^{(k)}(t)$	$\sigma^2(2k{-}1)!!/\ell^{2k}$	$\sigma^2\ell^{-2k}$

The double factorial and ℓ^{-2k} growth means higher-order derivative uncertainty explodes, especially for short lengthscales.

7.2 Posterior variance is smaller, but follows the same trend

Conditioning on data reduces each variance by the $K_{*n}\tilde{K}^{-1}K_{*n}^\top$ term. But the reduction is less effective for higher derivatives: the cross-covariance vectors $k^{(p,0)}(t_* - t_j)$ oscillate more rapidly and the effective number of informative training points decreases.

As a rough rule: if the data reduce the function-value variance by a factor of 100 (from prior σ^2 to posterior $\sigma^2/100$), they might only reduce g''' variance by a factor of 5–20, depending on data density relative to ℓ.

7.3 Consequence for the ratio

In (9), the denominator term $\mu_u^2\sigma_{vv}/\mu_v^4$ involves σ_{vv} (order-3 variance), which can be 1–2 orders of magnitude larger than σ_{uu} (order-2 variance). This makes the uncertainty in g''' the dominant contributor to $\mathrm{Var}(a)$ in most practical cases.

8 Step 7: A Complete Numerical Example

8.1 Setting

Consider a GP with hyperparameters $\sigma^2 = 1$, $\ell = 1$, $\sigma_n^2 = 0.01$, fitted to $N = 50$ equally spaced points on $[0, 5]$.

At $t_* = 2.5$ (center of the domain), suppose the posterior gives:

$$\mu_u = g''(2.5) = -0.8,$$
$$\mu_v = g'''(2.5) = 0.4.$$

So $\hat{a} = -0.8/0.4 = -2.0$.

The posterior covariance block (from (5)) is:

$$\Sigma_z = \begin{bmatrix} 0.020 & -0.005 \\ -0.005 & 0.180 \end{bmatrix}.$$

Note $\sigma_{vv} = 0.180 \gg \sigma_{uu} = 0.020$, consistent with the scaling discussion.

8.2 Computing $\mathrm{Var}(a)$

From (9):

$$\begin{aligned}
\mathrm{Var}(a) &= \frac{0.020}{0.4^2} - \frac{2(-0.8)(-0.005)}{0.4^3} + \frac{(-0.8)^2 \cdot 0.180}{0.4^4} \\
&= \frac{0.020}{0.16} - \frac{0.008}{0.064} + \frac{0.1152}{0.0256} \\
&= 0.125 - 0.125 + 4.500 \\
&= 4.500.
\end{aligned}$$

So $\sigma_a = \sqrt{4.500} \approx 2.12$, and a 95% confidence interval is

$$a \in \hat{a} \pm 1.96\,\sigma_a = -2.0 \pm 4.16 = [-6.16,\ 2.16].$$

8.3 Breakdown by term

Term	Interpretation	Value
σ_{uu}/μ_v^2	Numerator uncertainty	0.125
$-2\mu_u\sigma_{uv}/\mu_v^3$	Correlation correction	-0.125
$\mu_u^2\sigma_{vv}/\mu_v^4$	Denominator uncertainty	4.500
Total	$\mathrm{Var}(a)$	4.500

The denominator (order-3) term contributes over 97% of the total variance, with the correlation term providing some cancellation against the numerator term.

9 Summary

The complete chain from data to parameter confidence interval is:

1. **Fit GP** to noisy observations $\{(t_i, y_i)\}$, obtaining hyperparameters $(\hat{\sigma}^2, \hat{\ell}, \hat{\sigma}_n^2)$ and the Cholesky factor of $\tilde{K}$.

2. **Compute kernel derivative vectors** $k^{(p,0)}(t_* - t_j)$ for $p = 2, 3$ and $j = 1, \ldots, N$, using the Hermite recurrence (3).

3. **Form the posterior covariance** of (g'', g''') at t_* via

$$\Sigma_z = \Sigma_{\text{prior}}^{(2:3)} - K_{*n}^{(2:3)} \, \tilde{K}^{-1} \, (K_{*n}^{(2:3)})^\top.$$

4. **Compute the posterior means** μ_u, μ_v via (4), and the point estimate $\hat{a} = \mu_u/\mu_v$.

5. **Propagate uncertainty** via the delta method / IFT:

$$\text{Var}(a) = \frac{\sigma_{uu}}{\mu_v^2} - \frac{2\mu_u \, \sigma_{uv}}{\mu_v^3} + \frac{\mu_u^2 \, \sigma_{vv}}{\mu_v^4}.$$

6. **Form confidence interval**: $\hat{a} \pm z_{\alpha/2} \sqrt{\text{Var}(a)}$.

The dominant source of uncertainty is almost always the highest-order derivative in the denominator, because GP derivative variance grows as $\sigma^2 (2k-1)!!/\ell^{2k}$ with derivative order k, and conditioning on data provides diminishing reduction for higher orders.